

BERKELEY MFE x CAPITOLIS INDUSTRY PROJECT

Monte Carlo CCR Engine

Executive summary: methods, formulas, variance reduction, calibration, and results

18

reporting nodes

39

risk factors

6

real bugs fixed

2

real Hull-White calibrations

What This Covers

A companion to the full technical report -- same findings, visual-first

#	Topic	Covers
1	Foundations	Data collection recap, pricer contract, trade inventory
2	Simulation engine	Hull-White rates, correlated GBM equities/FX
3	Speed & variance reduction	Two real speedups; 5 RNG methods compared, two ways
4	Convergence, 95th and 99.9th percentile	How many paths are actually enough
5	Bloomberg data upgrade	Real swaption calibration, real JPY curve, real FX forwards
6	Greeks & MPOR	Pathwise vs bump-reprice; margin period of risk exposure

Rate Model: Hull-White One-Factor

Gaussian, mean-reverting, exact closed-form bond prices at every simulated node

$dr(t) = [\theta(t) - a \cdot r(t)] dt + \sigma dW(t)$, via the shifted-OU decomposition: $r(t) = x(t) + \alpha(t)$, where $\alpha(t)$ fits today's REAL Databento curve exactly at $t=0$ (verified: matches to $1e-9$ relative tolerance).

Parameter	Value	Source
sigma (vol)	0.63%	Realized SOFR spot vol -- real data
a (mean reversion)	0.0167 (was 0.0458)	REAL swaption vol cube (Bloomberg), preferred over the earlier SOFR-futures proxy

No remaining disclosed-guess parameters in the rate model -- both inputs now come from real, cross-validated market data.

Speed Optimization

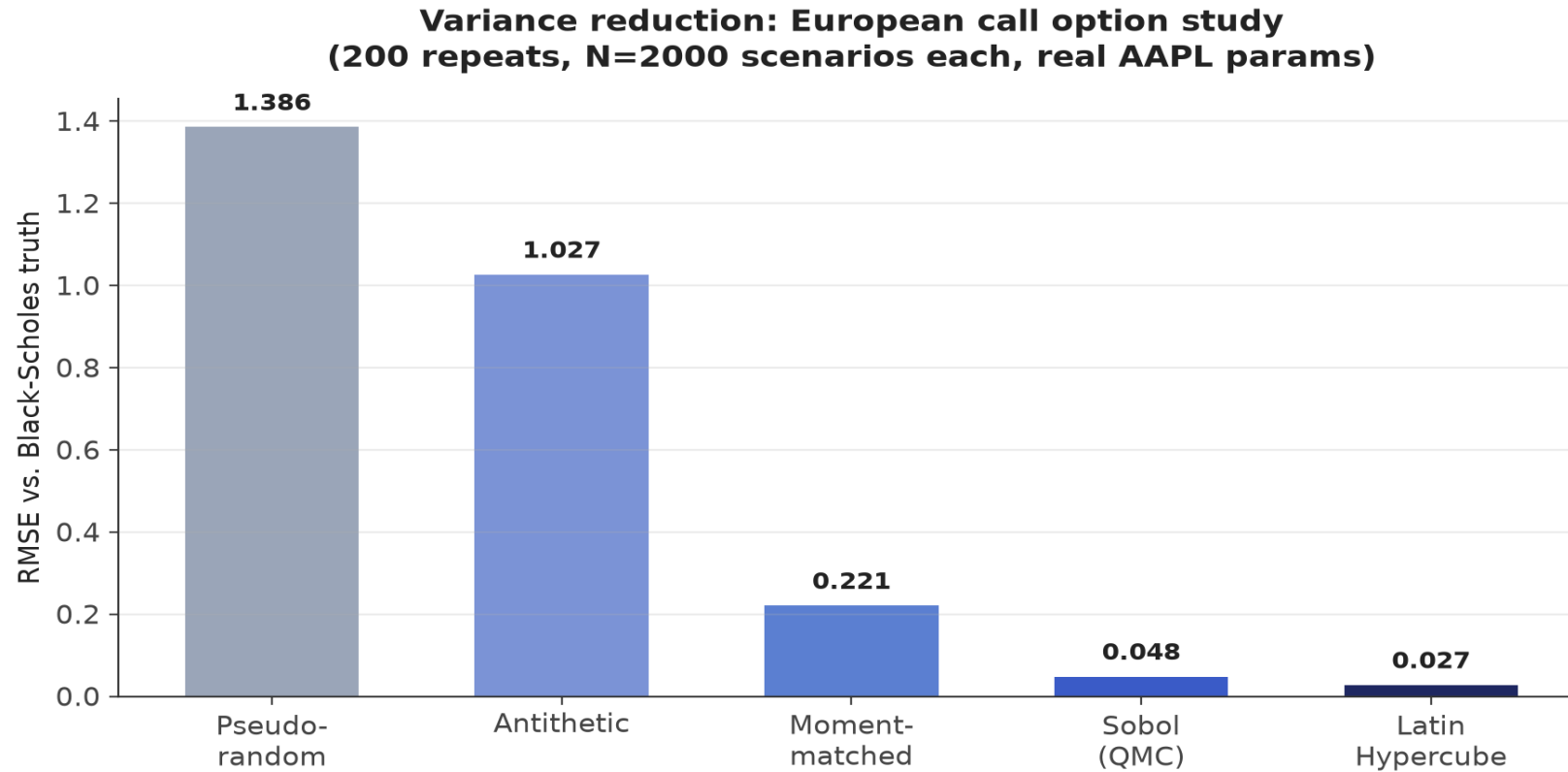
Two real bottlenecks found and fixed -- measured, not estimated

Fix	Before	After	Gain
Exact analytic curve (replaces fixed-pillar interpolation)	0.206 ms/call	0.019 ms/call	10.6x faster, more accurate
Fixed a Windows multiprocessing import bug	113 ms/scenario	51 ms/scenario	2.2x faster

Combined: full 16-trade, 2000-scenario simulation: ~12 min (projected) -> 93 seconds (~7.7x).

Variance Reduction: Controlled Option Test

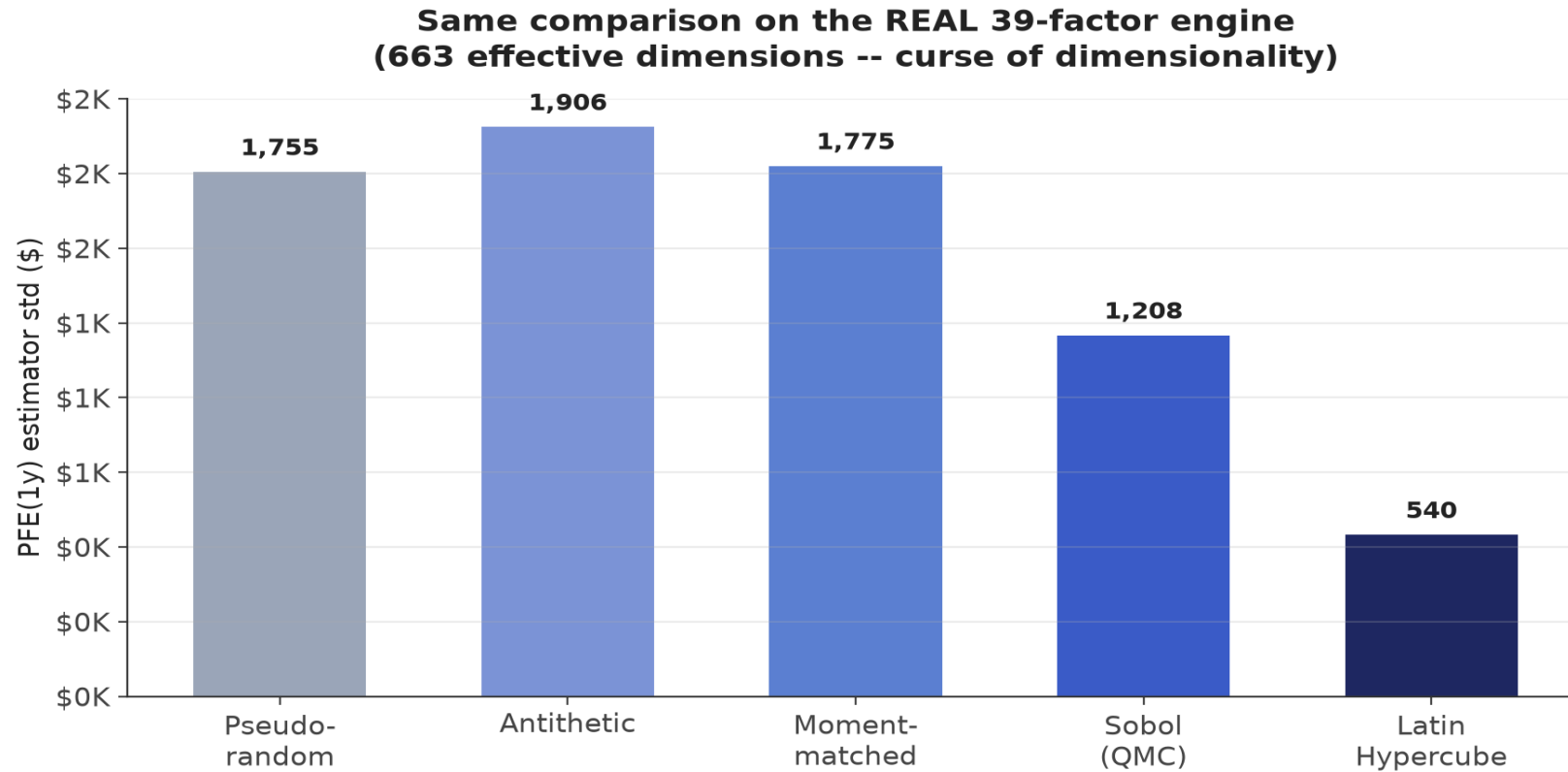
European call, real AAPL spot/vol, known Black-Scholes truth -- 5 methods compared



Latin Hypercube: ~50x lower RMSE than pseudo-random, at under 4x the per-trial cost.

Does It Hold on the Real Engine?

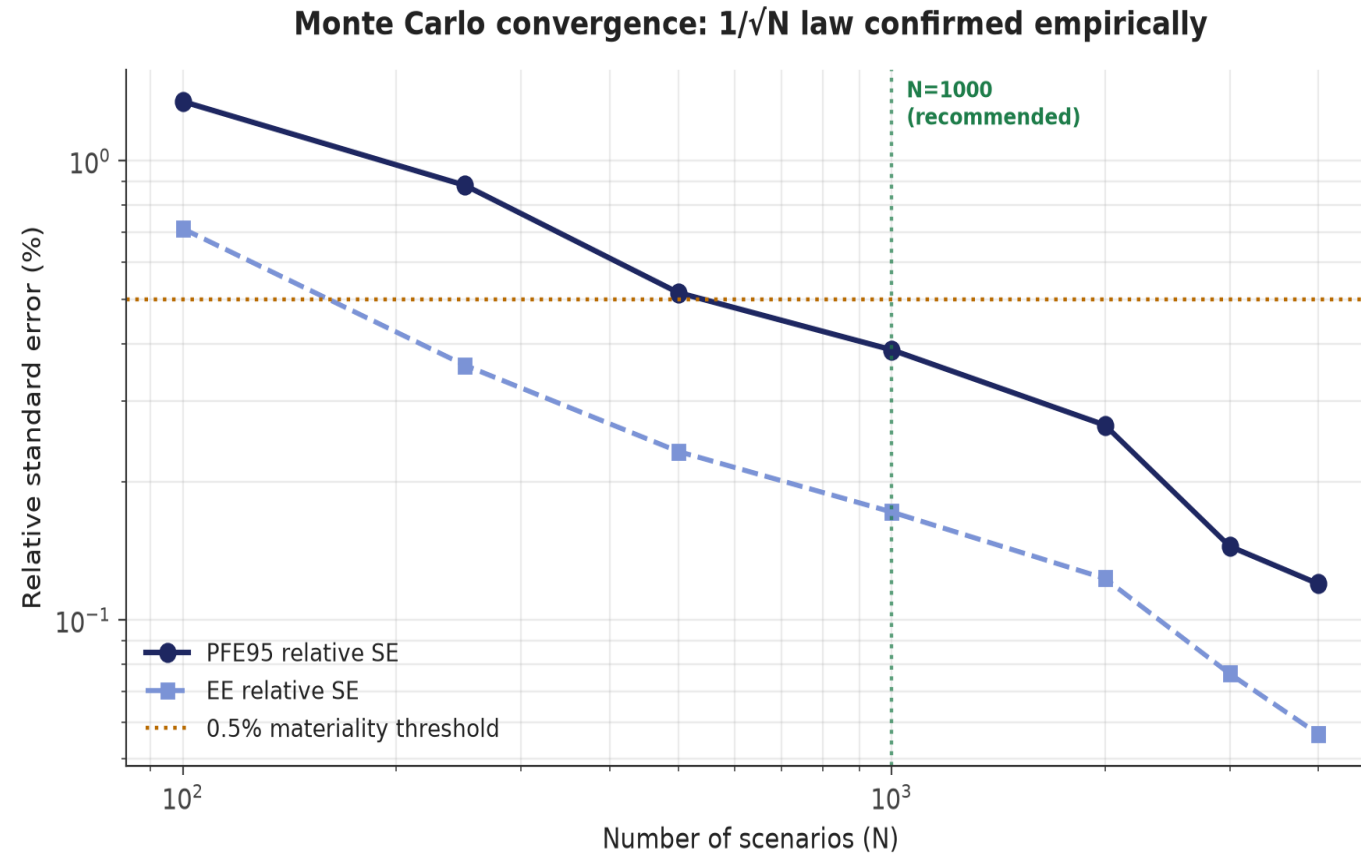
39 factors x 17 time steps = 663 effective dimensions



Antithetic/moment-matching show NO benefit at this dimensionality -- Latin Hypercube still wins (3.25x).

How Many Paths Are Enough? (95th percentile)

Empirical convergence study, bootstrap-resampled from a 6,000-scenario pool

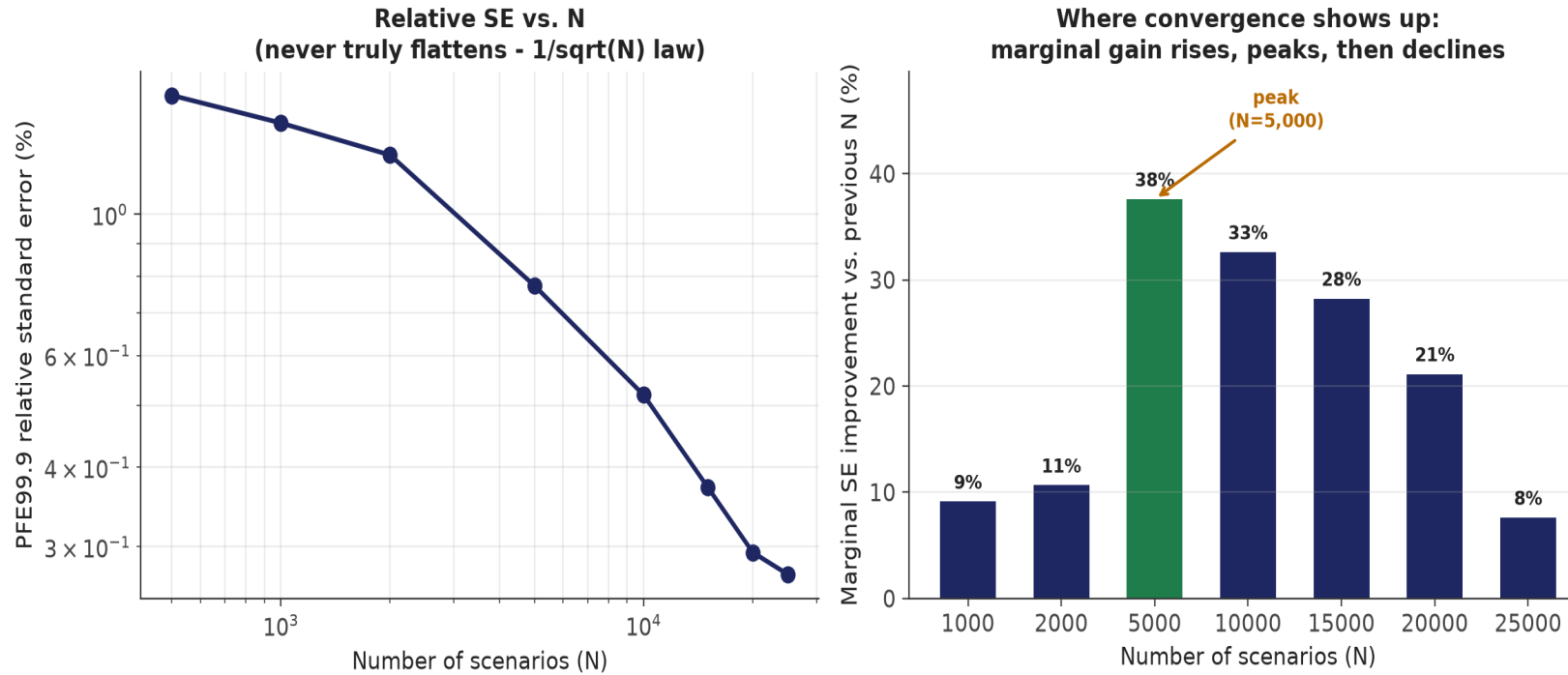


N=1,000 for iteration (0.39% error, 73s); N=2,000-3,000 for reporting (0.1-0.3%, 2-3 min).

...And at the 99.9th Percentile?

A materially harder problem -- only ~1 in 1,000 scenarios informs the tail. Extended to N=30,000.

PFE99.9 Tail Convergence Study (N up to 30,000, Latin Hypercube)

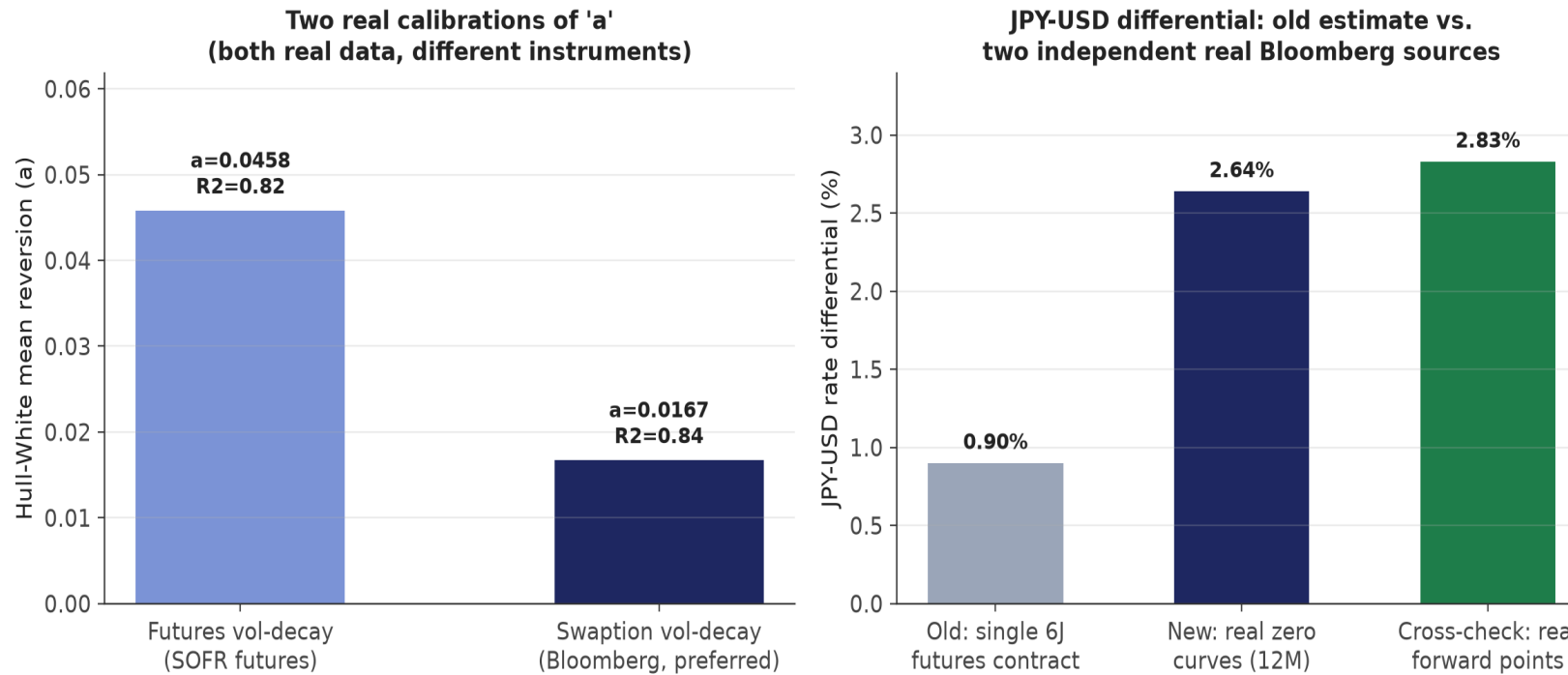


Marginal SE gain peaks at N=5,000, then declines. Recommendation: N=10,000-15,000, not 30,000.

Bloomberg Data Upgrade

Real swaption vols, real JPY curve, real FX forwards -- replacing three proxies

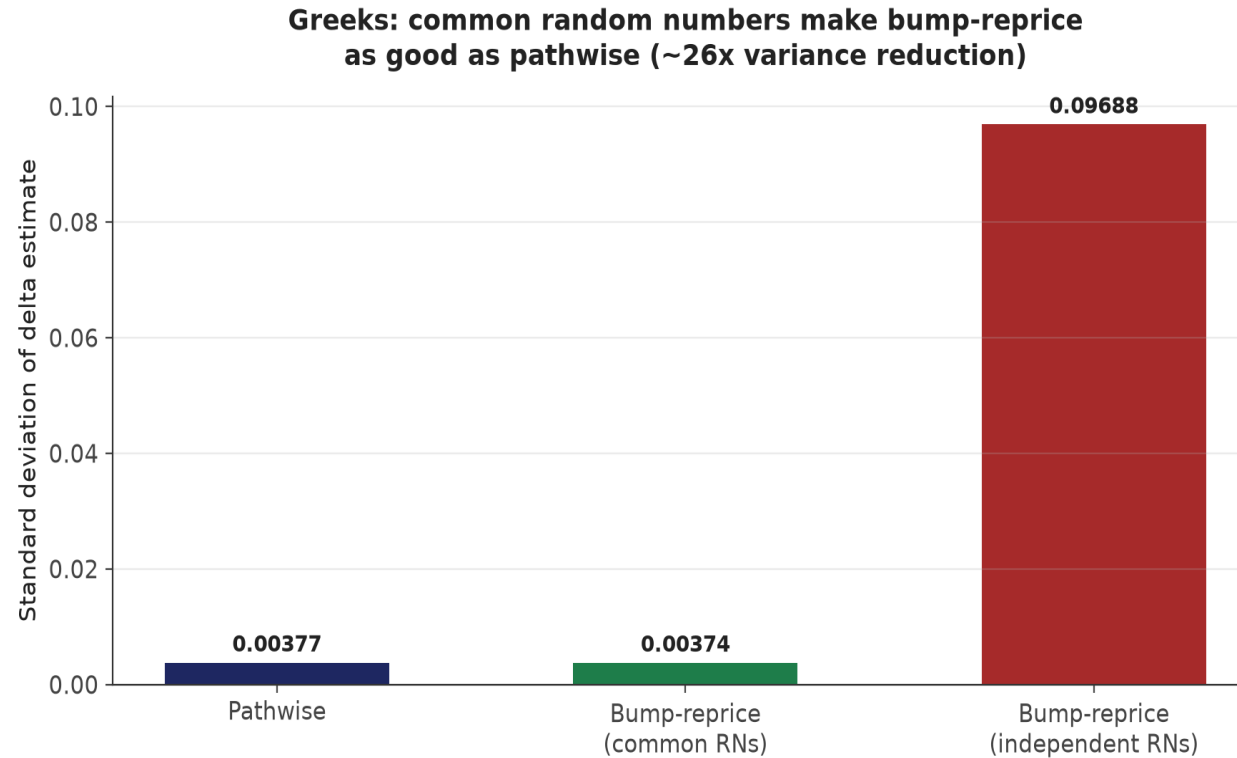
Bloomberg Data Upgrade: Two Real Calibrations Revisited



Two independent real sources (curves & forward points) agree closely: 2.64% and 2.83% JPY-USD differential -- correcting the old single-futures estimate of ~0.9%.

Extension to Greeks

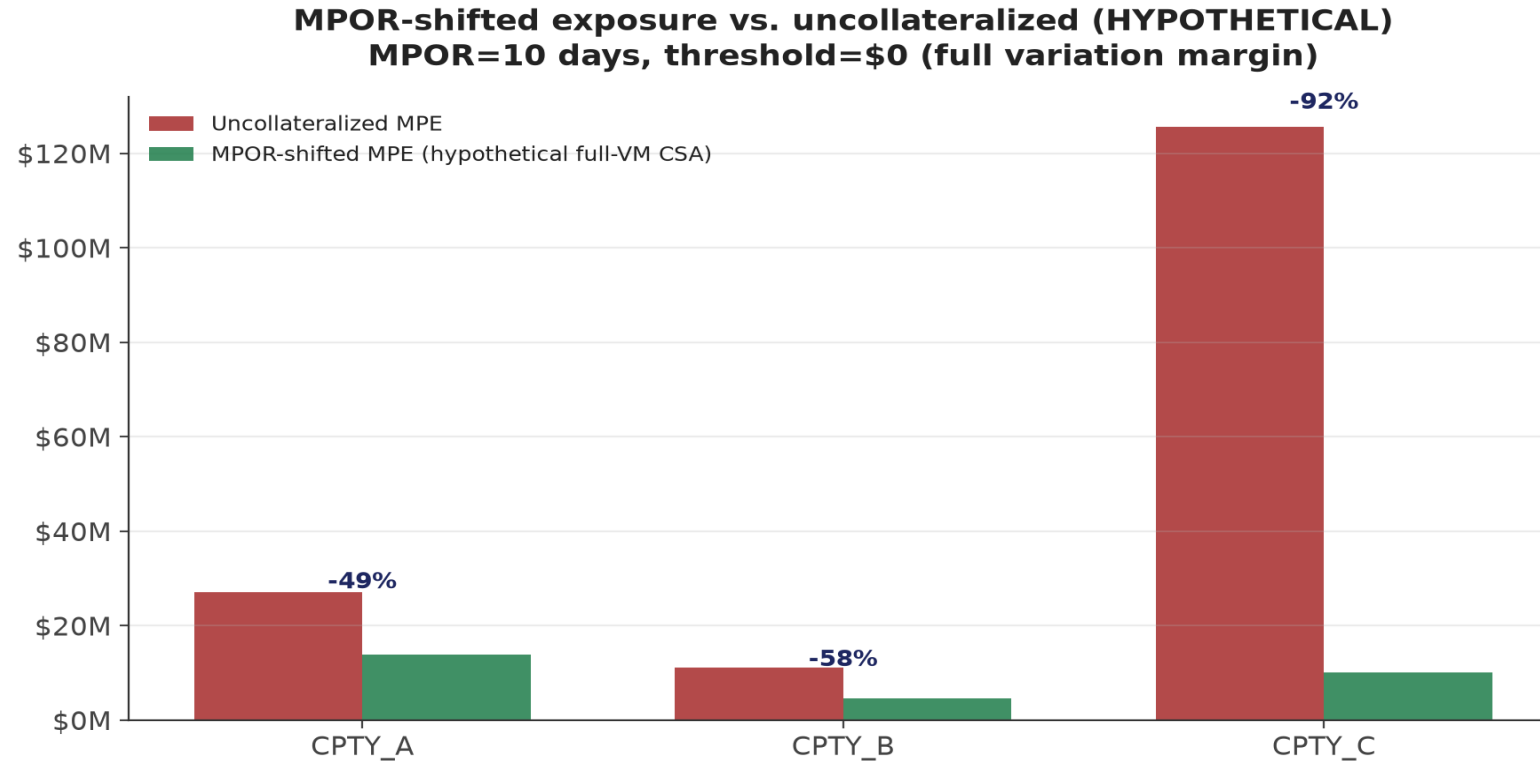
Pathwise vs. bump-and-reprice, with and without common random numbers



Common random numbers make bump-and-reprice as good as pathwise (~26x variance reduction vs. independent draws) -- the technique that generalizes to our real, black-box pricer library.

MPOR: Margin Period of Risk

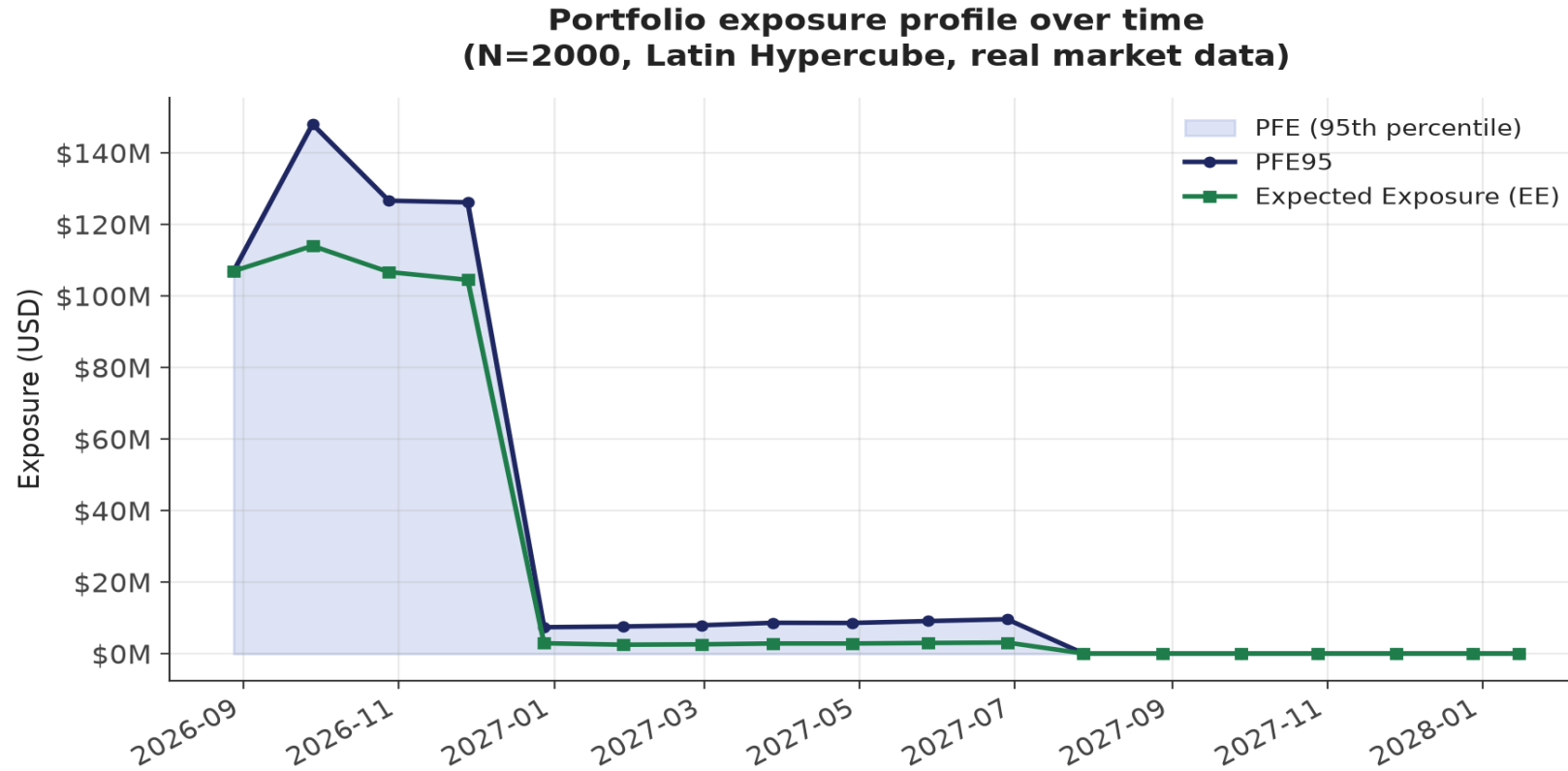
Was a 10-day MPOR used in production? No -- the book is uncollateralized. Built for when it applies.



Hypothetical CSA (threshold=\$0, MPOR=10 days): CPTY_C's exposure drops 91.9% once its \$500M bond forward's notional-driven MTM gets collateralized away.

Current Exposure & Exposure Profile

Real market data, self-validated to exactly 0.0000% internal consistency



Collapses >94% by end of Dec 2026 as most trades mature -- risk attention belongs in the next 4 months.

6 Real Bugs Found and Fixed

Every one caught by checking against something -- a second source, a full run, a hand calc -- not assumed

1	Ticker mismatch	BRK.B vs Yahoo's BRK-B, in two places
2	SOFR curve decade-wrap gap	Caught via Treasury cross-check
3	SOFR curve over-correction	Caught while calibrating Hull-White
4	US/Japan timestamp mismatch	~70% blank historical join, fixed
5	Stale self-validation false alarm	12.5hr-old file vs. real overnight moves
6	Curve-object mismatch	Now matches to exactly 0.0000%

Next Steps

#	Item
1	PFE-sensitivity Greeks: $d(\text{PFE})/dS$ at a future node, same bump-reprice + common-random-numbers mechanism
2	Confirm actual CSA/margin terms with Capitolis -- current book treated as uncollateralized by default
3	A full second stochastic JPY short-rate factor, using the real JPY OIS curve now in hand
4	Vectorized pricer reimplementation if scenario counts need to scale past ~10,000
5	SPX/TOPIX implied vols are index-level only -- not yet usable for our 37 individual equity names